Import Data Using Transforms Map (Spreadsheets)

Requirement Analysis, Data Flow Diagrams, and User Stories

3. Requirement Analysis

3.1 Functional Requirements

The Import Data Using Transforms Map feature allows users to import spreadsheet data into the system while applying predefined transformations. The system reads spreadsheet data, maps columns to system fields, validates values, transforms the data, and stores the final records.

- Users shall be able to upload spreadsheet files such as .xlsx or .csv.
- The system shall allow users to select a worksheet when a file contains multiple worksheets.
- The system shall identify spreadsheet headers and display them to the user.
- Users shall be able to map spreadsheet columns to destination fields.
- Users shall be able to create or select a transformation map.
- The system shall support renaming columns, changing data types, date formatting, text-to-number conversion, trimming spaces, case conversion, value replacement, combining or splitting columns, default values, and duplicate removal.
- The system shall validate required fields before importing data.
- The system shall identify invalid, missing, or incorrectly formatted values.
- The system shall display a preview of transformed data before final import.
- Users shall be able to confirm or cancel the import.
- The system shall save successfully transformed records in the appropriate data store.
- The system shall generate an import summary showing successful, failed, skipped, and duplicate records.

- The system shall provide error details that users can download or review.
- The system shall maintain an activity log for uploads, transformations, validations, and imports.
- Only authorized users shall be allowed to create, edit, execute, or delete transformation maps.

3.2 Non-Functional Requirements

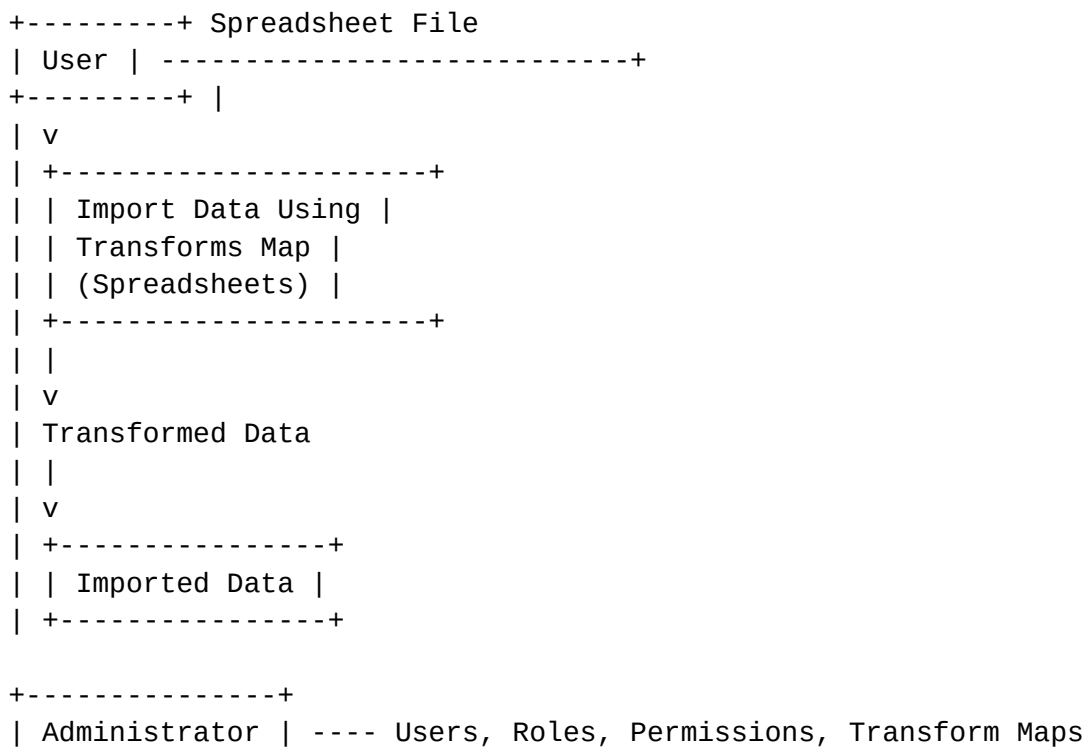
- The system should process spreadsheet files accurately without changing valid source values unexpectedly.
- The system should validate data before storing it and provide clear error messages.
- The system should support large spreadsheet files efficiently.
- The system should protect uploaded files and imported data from unauthorized access.
- The system should maintain data consistency when an import fails.
- The system should prevent partial imports unless explicitly permitted.
- The system should provide an easy-to-understand mapping and preview interface.
- The system should record the user, date, time, file name, transformation map, and import result.
- The system should allow approved transformation maps to be reused.
- The system should provide recovery when validation or import processing fails.

3.3 Data Flow Diagrams

Level 0 — Context Diagram

The system interacts with the User, Administrator, Spreadsheet File, and Notification or Reporting Service. Users upload spreadsheets, select maps, review previews, and confirm imports. Administrators manage users, permissions, and transformation maps.

Context-Level Data Flow



Level 1 — DFD

User → 1. Upload Spreadsheet → 2. Read Spreadsheet Structure → 3. Select/Create Transform Map → 4. Map Columns → 5. Apply Transformations → 6. Validate Data → Preview → 7. Confirm Import → 8. Store Imported Data → 9. Generate Import Summary → 10. Activity Log

Invalid data flows to Error Details and the Import Error Report. Valid data flows to the preview and then to the Imported Data Store after confirmation.

Main Data Stores

- User and Role Data
- Transformation Map Data
- Uploaded Spreadsheet Data
- Staging Data
- Imported Data
- Validation and Error Data
- Import History
- Activity and Access Logs

3.4 User Stories

User Story 1 — Upload Spreadsheet
As a user, I want to upload a spreadsheet file so that I can import its data into the system.

User Story 2 — Select Worksheet
As a user, I want to select a worksheet so that I can import data from the correct sheet.

User Story 3 — View Spreadsheet Headers
As a user, I want to view spreadsheet headers so that I can understand the available source columns.

User Story 4 — Select Transformation Map
As a user, I want to select an existing transformation map so that I can apply approved data-processing rules.

User Story 5 — Create Transformation Map
As a user, I want to create a transformation map so that spreadsheet columns can be converted into the required system format.

User Story 6 — Map Columns
As a user, I want to map spreadsheet columns to destination fields so that data is imported into the correct locations.

User Story 7 — Apply Data Transformations
As a user, I want to have the system transform spreadsheet values automatically so that imported data follows the required format.

User Story 8 — Validate Data
As a user, I want to have the system validate transformed data so that invalid records are detected before import.

User Story 9 — View Preview
As a user, I want to preview transformed records so that I can verify the results before saving them.

User Story 10 — Correct Mapping Errors
As a user, I want to correct mapping or transformation errors so that the import can be completed successfully.

User Story 11 — Confirm Import
As a user, I want to confirm the import after reviewing the preview so that only approved data is stored.

User Story 12 — Cancel Import
As a user, I want to cancel an import before confirmation so that incorrect data is not stored.

User Story 13 — View Import Summary
As a user, I want to view an import summary so that I know how many records succeeded, failed, were skipped, or were duplicated.

User Story 14 — Download Error Report
As a user, I want to download an error report so that I can correct invalid spreadsheet records.

User Story 15 — Manage Transformation Maps
As a user, I want to edit, approve, disable, or delete transformation maps so that only valid maps are available.

User Story 16 — Role-Based Access
As a user, I want to assign permissions for spreadsheet imports and transformation maps so that users access only permitted functions.

User Story 17 — Prevent Unauthorized Import
As a user, I want to prevent unauthorized users from uploading or importing spreadsheets so that organizational data remains protected.

User Story 18 — Maintain Import History
As a user, I want to view import history so that I can track when files were imported and which maps were used.

User Story 19 — Activity Logging
As a user, I want to view activity logs so that I can monitor spreadsheet uploads, transformations, imports, failures, and cancellations.

User Story 20 — Retry Failed Import
As a user, I want to retry a failed import after correcting errors so that valid data can be imported successfully.

Access Control

Role	Permissions
Administrator	Manage users, roles, transformation maps, permissions, logs, and all imports.
Import Manager	Upload spreadsheets, create or edit approved maps, validate data, and confirm imports.

Authorized User	Upload files and use assigned transformation maps.
Viewer	View import history, previews, and reports without changing data.
Unauthenticated User	No access to import functions or spreadsheet data.

Import Process Summary

1. The user logs in to the system.
2. The user uploads a spreadsheet.
3. The system reads the worksheet and displays its headers.
4. The user selects or creates a transformation map.
5. The user maps source columns to destination fields.
6. The system applies configured transformations.
7. The system validates required fields, data types, and business rules.
8. The user reviews the transformed-data preview.
9. The user corrects errors or confirms the import.
10. The system stores valid records and generates an import summary.
11. The system stores failed-record details in an error report.
12. The system records the complete operation in the activity log.